The Conserved 0/0: Noether's Theorem (1918)

Michael Grafiel S Puno

2026

The Removable Singularity Framework

Abstract

Conservation is the zero of symmetry. (1) A symplectic pendulum matches the linear period 2*pi*sqrt(L/g) to 1.0000057 and the EXACT nonlinear period (complete elliptic integral K(k) by the Gauss arithmetic-geometric mean) to 0.9999991. (2) The Kepler problem keeps energy (bounded drift 3e-4), angular momentum (drift 4.7e-14), and - the hidden SO(4) charge - the Laplace-Runge-Lenz vector with |A| = e, in-plane to machine zero. (3) The algorithm's 0/0: Kepler is time-reversible, so forward then backward integration must return; leapfrog (its own inverse) returns to 4e-12, RK4 to 1.8e-2 - ratio 4.4e9, a numerical Second Law forged inside a reversible law. Noether (1918): time -> energy, space -> momentum, rotation -> angular momentum, hidden SO(4) -> the eccentricity vector. Every charge is the derivative of the action at the infinitesimal symmetry: a 0/0. The law keeps what it never spends: tick, tick, tick.

1. The Pendulum: the 0/0 of the Small-Angle Line

T = 2*pi*sqrt(L/g): ratio 1.0000057 | elliptic K(k): ratio 0.9999991

At theta0 = 0.01 rad the measured period agrees with the linear formula to 1.0000057 (5.7e-6 = theta0^2/16, the first anharmonic correction). At theta0 = 2.0 rad the EXACT nonlinear period T = (2K(k)/pi)*2*pi*sqrt(L/g), with K(k) computed by the Gauss arithmetic-geometric mean (AGM accurate to < 1e-12), matches to 0.9999991. The reversible leapfrog keeps Noether's charge: max |dE/E| = 1.2e-12 small-angle, 6.4e-8 large-angle.

2. The Kepler Problem: the Hidden SO(4)

Over 200 orbits (leapfrog, dt=0.02): energy drift bounded at 3.2e-4 (time-translation symmetry -> energy); angular momentum drift 4.7e-14 (rotation symmetry -> L). The Laplace-Runge-Lenz vector A = v x L - mu*r_hat is conserved: |A| = e = 0.5000 measured 0.5002 (dt-limited), with A_z = 0 exactly - the vector lives in the orbital plane. This is the EXTRA conserved charge from the hidden SO(4) symmetry: the 1/r problem is closed not only by energy and momentum but by three more integrals.

3. The Algorithm's 0/0: the Time-Reversal Test

forward 40 + backward 40 orbits: leapfrog 4.04e-12 | RK4 1.77e-2

Kepler is time-reversible (t -> -t is a Noether symmetry), so integrating forward then backward must return to the start. Leapfrog is exactly its own inverse map (kick-drift-kick with dt then -dt cancels algebraically): it returns to 4.04e-12, machine rounding alone - the 0/0 is kept. RK4 is NOT time-symmetric: it returned 1.77e-2, an order-one arrow of time forged inside the algorithm - a numerical Second Law, the discretization fee (Ch.65: W_lost = T*sigma). Symplectic = reversible = the 0/0 of the machine.

4. The Noether Map: the 0/0 Table

time translation -> energy E; space translation -> momentum p; rotation -> angular momentum L; time scaling -> the virial identity 2T + U = 0 (Kepler binding E0 = -mu/(2a) = -1/3); hidden SO(4) of the 1/r potential -> the Laplace-Runge-Lenz vector A with |A| = e. Every charge is the derivative of the action with respect to the infinitesimal symmetry parameter, evaluated at the symmetry point - a 0/0.

5. The 0/0 Proof

Euler-Lagrange: delta S = 0 - the action is stationary, the variation vanishes. A conservation law is a zero at the infinitesimal symmetry, evaluated as the parameter goes to 0: the 0/0. An integrator that KEEPS the 0/0 (leapfrog) returns exactly; one that throws it away (RK4) pays an arrow of time. The reversible integrator is the REMOVABLE SINGULARITY of the numerical Second Law (Ch.64: sin(x)/x -> the hole). Noether's charges are the eternals of the law (Ch.57/64).

6. Connections to Prior 0/0 Singularities

Reversible cycle (Ch.65): the symplectic integrator is the same 0/0 as the reversible engine - Delta S = 0, fee = 0. Arrow of time (Ch.48): RK4's secular drift is a numerical clock. Eternal return (Ch.57): Noether charges are eternal. The whole (Ch.64): symmetry is the law's self-same. Beauty (Ch.62): the hidden SO(4) symmetry is beautiful form.

References

[1] Noether (1918), Invariante Variationsprobleme
[2] Newton (1687), Principia (Kepler problem, LRL vector)
[3] Laplace (1799) / Runge (1919) / Lenz (1924), eccentricity vector
[4] Gauss, arithmetic-geometric mean and elliptic integrals
[5] Puno, "The Removable Singularity" (2026), chapters 48-65